STAT 312: HW Assignment 5

Instructor: Dr. Md Mutasim Billah

Submission Guidelines

Please follow these guidelines carefully when completing and submitting your homework. Clear and well-organized work helps ensure fair and accurate grading.

- There are **9 problems**.
- Submit all homework assignments through **Canvas**.
- If you prefer to work with **pen and pencil**:
  - Download and/or print the homework assignment.
  - Scan your completed work and combine all pages into a **single PDF file**.
  - Upload the PDF to the corresponding homework assignment on Canvas.
- If you prefer to use a **writing device** or type your solutions, you may complete the homework electronically and submit your final work directly as a **PDF**.
- You may also need to submit an **R file** if the homework includes a programming task.
- Your work must be **clear, organized, and legible**. Please follow these presentation guidelines:
  - Clearly circle or otherwise mark your final answers so they can be easily located on the page.
  - Show all necessary work for **nontrivial calculations**. Answers without sufficient supporting work may receive little or no credit.
- Name your submitted PDF using the following format:

  `firstname_lastname_HW#.pdf`

  For example: `john_smith_HW#.pdf`

Problem 1

What are the degrees of freedom corresponding to each of the following probabilities?

Instruction: Draw an appropriate sketch for each part.

(a) $P(T > 2.763) = 0.005$.

(b) $P(T < -2.650) = 0.01$.

(c) $P(-1.313 < T < 1.313) = 0.80$.

Problem 2

When computing an 81% confidence interval for μ when σ is known, using the formula

$$\overline{Y} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

what specific value should you use for $z_{\alpha/2}$?

Problem 3

Hundreds of cross-country skiers participate in a race during Winter Carnival weekend. A student is interested in μ, the population mean number of minutes needed to complete the race among all the skiers. Assume that the population is normally distributed. The population standard deviation is known to be $\sigma = 17.7$ minutes.

If the sample mean for a random sample of 16 skiers is 51.8 minutes, what is the 95% confidence interval for μ?

Problem 4

Jacob's confidence interval for μ was $(59.59, 72.41)$. The population standard deviation was known to be $\sigma = 50$, and the sample size was $n = 100$.

(a) What was the sample mean?

(b) What confidence level did Jacob choose?

(c) Suppose Jacob wanted a margin of error no larger than 3 at the same confidence level. What sample size would be necessary?

Problem 5

When computing a 99% confidence interval for μ when σ is unknown and $n = 10$, using the formula

$$\overline{Y} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},$$

what specific value should you use for $t_{\alpha/2,n-1}$?

Problem 6

An agricultural scientist is interested in a population of chicken eggs, particularly in μ, the population mean weight of the eggs in grams. Assume that the population is normally distributed.

A random sample of 13 eggs has a sample mean of 57.7 grams and a sample standard deviation of 4.2 grams. The population standard deviation σ is unknown. What is the 95% confidence interval for μ?

Problem 7

Is America's romance with movies on the wane? In a Gallup poll of $n = 800$ randomly chosen adults, 45% indicated that movies were getting better, whereas 43% indicated that movies were getting worse.

(a) Find a 98% confidence interval for p, the overall proportion of adults who say that movies are getting better.

(b) Does the interval include $p = 0.50$? Based on your confidence interval, is there sufficient evidence to conclude that a majority of adults say that movies are getting better? Explain your reasoning.

Problem 8

A book publishing company has recently produced a batch of several thousand copies of a hardcover book. The company has hired a team of engineers to inspect its machines because too many books have been assembled with missing pages.

Let p denote the proportion of defective books in the batch. Of 160 randomly selected books, 96 are defective. The company would like a 90% confidence interval for p with a margin of error no larger than 0.01.

What sample size is necessary? Use the estimate of p from the original sample.

Note: Round the required sample size up to the next whole number.

Problem 9

Refer to Problem 8. Suppose the book publishing company had no prior estimate of p. What sample size would be necessary to achieve a margin of error no larger than 0.01 with 90% confidence?

Note: Round the required sample size up to the next whole number.